

AI DISCOVERERS

Pattern Detectives

How Machines Notice Similarities

Classroom slides

Grades 3–5

45–60 minutes

Notice a pattern. Explain your rule.



Today's Learning Goals

WE WILL

- 01 Notice similarities and differences.
- 02 Sort objects using a clear rule.
- 03 Compare different sorting rules.
- 04 Explain how AI uses examples.
- 05 Check for mistakes or unfair groups.

I CAN SHOW MY LEARNING

- 01 Explain the rule I chose.
- 02 Use evidence for each choice.
- 03 Explain another rule that could work.
- 04 Say why people check AI results.

Mystery Sorting

Warm-up



LOOK AT THE TWO GROUPS

Your teacher sorts these objects without telling you the rule.

What do the objects in each group have in common?

Could another rule also work?

Use an object as evidence for your idea.

What Is a Pattern?

A pattern is a feature, order, or relationship we notice.

Patterns can repeat or help us organize ideas.

FEATURES WE CAN USE

Color

Size

Shape

Material

Use

Location

TALK WITH A PARTNER

Name one way to sort school objects and one way to sort foods.

Pattern Detective Challenge

Activity 1 · student lesson p. 3

SORT THE SAME EIGHT OBJECTS INTO GROUP A OR GROUP B

- Backpack
- Basketball
- Book
- Bicycle
- Tablet
- Tennis shoe
- Water bottle
- Paintbrush

Use what an object is made of, what it does, or another clear feature. Do not use color alone. You can write, point, or sort real objects.

YOUR GROUP'S TASK

- 01 Choose a rule you can explain.
- 02 Sort the eight objects into Group A or Group B.
- 03 Record the group for each object.
- 04 Explain the feature your rule uses.
- 05 Compare with another group's rule.

COMPARE YOUR THINKING

- Which feature mattered?
- Which object was hardest to sort? Why?
- What did others notice?

Two rules can fit the same objects.

How Machines Learn From Examples

- 01 Some AI systems find patterns in many examples and use them to make predictions.
- 02 The examples and their labels affect the result.
- 03 Missing examples, wrong labels, or a poor rule can lead to mistakes.
- 04 **People check the result and decide what to do.**



A rule might use color or shape. Where would the red circle go?

Our sorting activity models grouping. It does not train an AI system.

Examples and Labels

Activity 3 · student lesson p. 5

IMAGINE A GROUP CALLED “USEFUL FOR LEARNING.”

Choose a label for each object. Explain the learning activity you have in mind.

Book	YES / NO	Pencil	YES / NO
Video game controller	YES / NO	Magnifying glass	YES / NO
Notebook	YES / NO	Headphones	YES / NO
Soccer ball	YES / NO	Map	YES / NO

Why did you choose those labels? Could a partner choose differently?

MISSING EXAMPLES CAN MISLEAD

Imagine the only examples labeled “useful for learning” are books, notebooks, and maps.

A pretend model notices paper and guesses headphones are not useful. That guess misses learning by listening.

What example or information is missing? What should a person check?

A model, not a real AI tool. This is a simplified simulation, not an actual AI prediction.

Design a Sorting Rule

Activity 2 · student lesson p. 4

MAKE A NEW RULE

- 01 Choose a different feature.
- 02 Sort the same examples again.
- 03 Write or explain how your rule works.
- 04 Ask a partner to test your rule.

YOUR PARTNER CHECKS

- Can I follow the rule?
- Which example is hardest to place?
- Could one object fit in both groups?

Use the feedback to improve your rule.

Human Judgment Check

Activity 4 · student lesson p. 6

01 An unfair guess

A pretend machine labels every student wearing glasses as “likes reading.” Glasses do not tell us a person's interests.

02 A difficult image

A photo system puts an object in the wrong group because the photo is dark.

03 A narrow recommendation

A library app suggests only books like the first book a student borrowed.



A visible feature does not tell us a person's interests.

For each one: What is missing or wrong? What should a person check or change?

These are fictional scenarios, not claims about a specific tool.

The Future Ready AI Learning Cycle

01 Launch

Start with a question or challenge.

02 Explore

Investigate examples and ideas.

03 Explain

Build understanding together.

04 Create

Apply learning in a task.

05 Reflect

Think about choices and results.

06 Extend

Connect learning to another situation.

Today we tried each step by noticing, sorting, testing, and checking.

Exit Ticket

Student lesson p. 7

MACHINE PATTERNS

A machine can notice patterns by...

.....

.....

HUMAN ROLE

A person should check the result because...

.....

.....

MY TAKEAWAY

One thing I learned about patterns is...

.....

.....

MY QUESTION

One question I still have is...

.....

.....

Write, draw, or explain your answers aloud.

My Responsible AI Promise

Student lesson p. 7

I WILL

- ☐ Protect private information.
- ☐ Check important information.
- ☐ Notice mistakes or unfair groupings.
- ☐ Use technology to support my own thinking.
- ☐ Ask a trusted adult when I am unsure.

